

18.06 Linear Algebra — Lectures 20, 21, 22
Textbook Notes — Sections 5.3, 6.1, 6.2

• Section 5.3 Cramer's Rule, Inverses, Volumes

• Every entry of A^{-1} and every component $A^{-1}b$ is a ratio of two determinants

• These formula's are computationally impractical (Cramer's Rule) for large matrices but provide symbolic expressions and connect determinants to geometry.

• Cramer's Rule

Solves $Ax = b$ component by component.
Each component x_j is a ratio:

$$x_j = \frac{\det B_j}{\det A}$$

↳ $B_j \rightarrow$ matrix A with column j replaced by b .

↳ Requires $n+1$ determinant evaluations (one for A , one per component).

• Inverse by Cofactors

$$A^{-1} = C^T / \det A$$

• The (i, j) entry of A^{-1} is the C_{ji} (note: ji not ij) divided by $\det A$.

• Cofactor matrix C has entries: $C_{ij} = (-1)^{(i+j)} \det M_{ij}$

→ Verification: $\{AC^T = (\det A)I\}$

- ↳ On the diagonal, row i of A dots with cofactors from row i (cofactor expansion), giving $\det A$.
- ↳ Off the diagonal, row i dots with row cofactors from different rows, giving zero.

• Volumes

The absolute value of the determinant gives the volume of the box (parallelepiped) whose edges are rows (or columns) of A .

$$\boxed{\text{volume} = |\det A|}$$

- ↳ For 2×2 : area of parallelogram with sides (a, b) and (c, d)

$$\text{area} = |ad - bc|$$

- ↳ For triangle: area of triangle with corners (x_1, y_1) , (x_2, y_2) , (x_3, y_3) :

$$\text{area} = \frac{1}{2} \left| \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} \right|$$

• Cross Product (3D only)

Properties

- ↳ $u \times v$ is perpendicular to both u and v
- ↳ $v \times u = -(u \times v)$
- ↳ $\|u \times v\| = \|u\| \|v\| |\sin \theta| = \text{area of parallelogram with sides } u \text{ and } v$
- ↳ $u \times u = 0$.

$$\underline{CP}: \left\{ u \times v = \det \begin{bmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix} \right\}$$

◦ Section 6.1 Eigenvalues

An eigenvector $\underline{x}$ is a special direction that $\underline{A}$ does not rotate (multiplication by $\underline{A}$ only stretches or shrinks it by a factor λ).

Finding these directions and factors reduces matrix problems to scalar problems.

• Definition : Eigenvalues and Eigenvectors

$$\boxed{Ax = \lambda x} \quad (x \neq 0)$$

~~λ~~ λ is the eigenvalue, x is the eigenvector.

Any nonzero scalar multiple of x is also an eigenvector.

Method to find eigenvalues & eigenvectors

- ① Form $A - \lambda I$ and solve $\det(A - \lambda I) = 0$
 - ↳ a degree- n polynomial in λ (the characteristic polynomial)
 - ↳ Its n roots are the eigenvalues

- ② For each λ , solve $(A - \lambda I)x = 0$ to find the eigenvector(s).

For 2×2 : $\det(A - \lambda I) = \lambda^2 - (\text{trace})\lambda + \det A = 0$

↳ solved by quadratic formula.

• Quick Checks

$$\lambda_1 + \lambda_2 + \dots + \lambda_n = \text{trace}(A) = a_{11} + a_{22} + \dots + a_{nn}$$

$$\lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n = \det A$$

• Eigenvalue Inheritance

If $Ax = \lambda x$, then :

$$\left\{ \begin{array}{l} A^2 x = \lambda^2 x \\ \rightarrow A^k x = \lambda^k x \\ A^{-1} x = \lambda^{-1} x \quad (\text{if } A \text{ invertible}) \\ (A + cI)x = (\lambda + c)x \end{array} \right.$$

↳ Eigenvectors do not change (only eigenvalues transform)

• Special Matrices

• Singular $\rightarrow \lambda = 0$ is an eigenvalue (nullspace vectors are eigenvectors for $\lambda = 0$)
↳ A ^{not} invertible

• Triangular $\rightarrow$ eigenvalues are the diagonal entries

• Projection $\rightarrow \lambda = 0$ and $\lambda = 1$ only.

• Reflection by 90° $\rightarrow \lambda = i$ and $\lambda = -i$ (complex)

• Symmetric $\rightarrow$ all λ real, eigenvectors perpendicular

• Warnings

- ① Eigenvalues of AB are generally not products of eigenvalues of A and B (unless they share eigenvectors)
- ② Eigenvalues of $A + B$ are generally not sums of eigenvalues of A and B
- ③ Elimination changes eigenvalues. $\rightarrow$ pivots of U are not eigenvectors of A .

Diagonalizable? $\rightarrow$ no duplicate eigenvalues?
 $\hookrightarrow$ the certainly yes
 $\hookrightarrow$ duplicates
 $\hookrightarrow$ check if eigenvectors span space.

Missing / Incomplete Eigenvector set?
 $\hookrightarrow$ Repeated Eigenvalue.

• Section 6.2 Diagonalizing a Matrix

If A has n independent eigenvectors, it can be factored as $A = X \Lambda X^{-1} = S \Lambda S^{-1}$, where Λ is diagonal.

|| Makes powers, difference equations, and matrix analysis trivial
↳ everything reduces to powers of scalars.

• The diagonalization

Place the n independent eigenvectors as columns of X ,
eigenvalues on the diagonal of Λ :

$$AX = X\Lambda \rightarrow A = X\Lambda X^{-1} \rightarrow \Lambda = X^{-1}AX$$

↳ Works because column k of AX is $AX_k = \lambda_k X_k$,
which matches column k of $X\Lambda$.

• Powers of A :

$$A^k = X \Lambda^k X^{-1}$$

↳ Powers of a diagonal matrix are trivial
 $\Lambda^k = \text{diag}(\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k)$.

• Solving Difference Equations $u_{k+1} = A u_k$

The solution $u_k = A^k u_0$ is computed in three steps:

1. Decompose: write $u_0 = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$, by

solving $c = X^{-1} u_0$

2. Multiply: each eigenvector gets scaled by λ_i^k

3. Sum

3. Recombine

$$u_k = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 + \dots + c_n \lambda_n^k x_n$$

↳ Separates a coupled system into n independent scalar sequences.

• When diagonalization works (and fails)

• Guaranteed : n distinct eigenvalues

- ↳ eigenvectors are automatically independent
- ↳ diagonalizable.

• Possible failure : repeated eigenvalues may produce too few independent eigenvectors.

- ↳ The X is not invertible & A cannot be diagonalized.

• Invertibility & Diagonalizability are independent properties :

- ↳ Invertibility depends on whether $\lambda = 0$

- ↳ Diagonalizability depends on whether there are sufficient eigenvectors.

• Similar Matrices

→ Matrices A and C are similar if $A = BCB^{-1}$ for some invertible B .

→ Similar matrices share the same eigenvalues (but generally different eigenvectors).

→ Diagonalizable matrices with the same Λ form one similarity class.

- Convergence

~~Matrices~~ $A^k \rightarrow 0$ if and only if all $|\lambda_i| < 1$.

If the largest $|\lambda|$ equals 1 (with eigenvector x_1),
then $A^k u_0$ approaches a multiple of x_1 (the "steady state").